Mohamed Ehab

Oracle Technical Consultant

Professional Summary

Oracle Technical Consultant specializing in AI-augmented enterprise applications, with expertise across Oracle APEX, Oracle SQL, and PL/SQL. Proven track record designing natural-language-to-SQL reporting assistants and multi-agent AI solutions on Oracle Fusion AI Agent Studio, alongside custom/standard report modification and OAF-based module enhancement. Combines strong PL/SQL and REST/JSON integration capabilities with applied AI expertise in RAG, LLM fine-tuning, and multi-agent orchestration, delivering end-to-end solutions across Oracle Cloud, Azure, and AWS.

Education

Zagazig University Egypt

B.Sc. Computer Science, AI Specialization — Faculty of Computers and Information Sciences

- Graduation project selected among the top 5 projects in the faculty by the academic committee.

Certifications

Agentic AI Certified Foundations Associate | Oracle 10/2026

Oracle Fusion AI Agent Studio Certified Foundations Associate — Rel 1 | Oracle [Certificate] 05/2026

Oracle APEX: Foundations | Coursera [Certificate] 07/2026

Oracle Cloud Infrastructure AI Foundations | Coursera [Certificate] 08/2026

OCI AI Foundations Associate (2025) | Oracle [Certificate] 05/2026

AWS Certified AI Practitioner | Amazon Web Services (AWS) via Manara [Certificate] 2026

Mastering Deep Learning | ScSkilled — Dr. Mostafa Saad Ibrahim [Certificate] 08/2024 – 03/2026

Machine Learning Diploma | ScSkilled — Dr. Mostafa Saad Ibrahim [Certificate] 07/2023 – 07/2024

CS224N: Natural Language Processing with Deep Learning | Stanford University (Self-Study)08/2025 – 01/2026

Technical & Soft Skills

Oracle Ecosystem (Fusion & EBS): Oracle Fusion AI Agent Studio, Oracle Fusion REST APIs, Oracle APEX , Oracle SQL, PL/SQL, Standard PL/SQL APIs, BI Publisher, OTBI, Oracle Cloud Infrastructure (OCI),Oracle Report Builder , Oracle Workflow Builder

Software Engineering & Languages: Python, Java, C++, JavaScript, FastAP, asyncpg, Pydantic, Streamlit, REST APIs, Data Structures, Algorithms, OOP, Design Patterns

AI, LLMs & NLP: LLM Fine-Tuning, LoRA, SFT, Knowledge Distillation, Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, vLLM, LangChain, Prompt Engineering, Arabic NLP, English NLP, PyTorch, scikit-learn, Hugging Face Transformers, LLaMA-Factory

Computer Vision: YOLOv11, YOLOv12, ResNet, Vision Transformer (ViT), Object Detection, Instance Segmentation, Edge Inference, OCR, OpenCV

Databases & Vector Stores: Oracle SQL ,PostgreSQL, Qdrant, PgVector, Alembic

Infrastructure & MLOps: Oracle Cloud Infrastructure (OCI) , Docker, GitHub Actions, CI/CD, Microsoft Azure, AWS, Prometheus, Grafana, Weights & Biases, Nginx, Locust

Knowledge Transfer: Organizes an industry-readiness study group for aspiring AI engineers, producing structured educational content and guiding members toward junior AI roles

Cross-Functional Delivery: Comfortable owning full project lifecycles — from requirements gathering through production rollout — across Oracle, AI, and cloud domains

Self-Directed Problem Solving: Diagnoses and resolves complex integration issues independently (e.g., Oracle HCM REST API troubleshooting, multi-agent orchestration design) with minimal guidance

Work Experience

DTCommittee
05/2025 – Present

Oracle Technical Consultant — *Giza, Egypt*

- Deliver AI-augmented Oracle APEX and Oracle Fusion solutions for enterprise business functions, owning schema design, PL/SQL development, REST API integration, and security hardening across the full delivery lifecycle .
- Engineered a AI AGENT assistant For EBS using Oracle Apex with multi-layer security guardrails, achieving **94%+ accuracy**, a **2s average response time**, and zero unsafe query executions.
- Integration with ETA (Egyptian Tax Authority) to automate supplier invoice creation. , and an Oracle analytics dashboard.
- Architected a multi-agent business process assistant on Oracle Fusion AI Agent Studio, integrating Oracle Fusion REST APIs with a retrieval-grounded policy Q&A agent.
- Build custom and Modify standard reports and enhanced custom modules built on OAF pages

OZ-Tech
02/2025 – 07/2025

Junior AI Engineer — Part-Time, Remote — *Egypt*

- Trained and deployed YOLOv12 object detection and instance segmentation models achieving **90%+ Precision** and **87% Recall** across multiple production environments.
- Owned the complete ML pipeline end-to-end: data collection, annotation, augmentation, model training, evaluation, and production deployment on real-world industrial datasets.
- Integrated deployed models with Arduino-based edge devices, enabling real-time on-device inference at **sub-100ms latency** in resource-constrained environments.
- Collaborated with cross-functional teams in an agile environment to deliver operational AI systems on schedule, supporting internal training programs in detection and segmentation tasks.

RMG
02/2025 – 05/2025

AI Engineer Intern — Egypt Office — *Saudi Arabia*

- Developed and shipped computer vision models for object detection and segmentation, driving projects from research prototype to live production delivery.
- Accelerated model iteration cycles by implementing structured evaluation pipelines, reducing review time and improving deployment readiness for production handoff.
- Collaborated with cross-functional AI teams to integrate computer vision models into operational workflows across the full project lifecycle.

Projects

AI Invoice Automation Agent — Oracle AI Agent Studio | *Infrarose* 2026

- Designed and built an AI workflow agent that automates supplier invoice intake for Oracle Fusion Cloud Payables: a user uploads a PDF invoice in a chat window, and the invoice is created and validated in Fusion in about one minute, with no manual data entry.
- Implemented PDF-to-JSON extraction with reconciliation checks between line amounts and the header total, and separate processing paths for PO and non-PO invoices.
- Integrated Fusion REST APIs for supplier, supplier site, PO header and PO line lookups, and for invoice creation and the `validateInvoice` action; managed agent state with persistent variables.
- Developed line matching by description and unit price instead of line order, and automated GL account classification for non-PO lines using a mapping table.
- Added human-in-the-loop checkpoints only for ambiguous supplier sites and final approval, keeping accountants in control of judgment calls.

AI-Powered NL-to-SQL Assistant — Oracle APEX | *DTCommittee* 2026

- Built an Oracle APEX tool letting non-technical HR users query the database in plain English; translated natural language into safe, read-only Oracle SQL via the Cohere Chat API (`command-a-03-2025`) and auto-executed it through `DBMS_SQL`.
- Constrained generation to a single explicitly defined HR schema, rejecting any request referencing unknown tables or columns, and rendered results as a live formatted HTML report inline in APEX.
- Implemented defense-in-depth security: prompt-level restriction to `SELECT`/`WITH` statements, prompt-injection guarding, and a server-side guard validating every generated statement's keyword and regex-scanning for DML/DDL/DCL before execution; all output escaped via `APEX_ESCAPE.HTML`.
- Reached **94%+ accuracy** on schema-valid SQL generation, a **2s average response time**, and zero unsafe query executions reaching the database.

Invoice Automation Portal — Oracle APEX | *DTCommittee* 2026

- Architected a full REST-to-database import pipeline pulling live invoices from an external e-invoicing API (Egyptian ETA-style) via Oracle APEX Web Source Modules and REST Data Sources, with credential-store authentication and duplicate-import protection.
- Parsed invoice header and line-item JSON with `APEX_JSON` into normalized Oracle tables, and built editable Interactive Grids with full CRUD and lost-update protection for vendor/finance review.
- Built an AJAX-driven, one-click invoice submission workflow (custom JavaScript dynamic actions triggering `APEX_INIT_WF`) and a hand-built, print-ready HTML/CSS invoice rendering engine.
- Designed an executive analytics dashboard with Oracle JET charts (trends, top vendors, top invoices, distribution by vendor) and dual authentication (native APEX accounts + custom EBS SSO).

Oracle E-Business Suite (EBS) Project | *Al-Babtain Power & Telecom* 2026

- Delivered custom and standard reports modification for Al-Babtain Power & Telecom on Oracle E-Business Suite (EBS).
- Updated and enhanced custom modules built on OAF (Oracle Application Framework) pages, ensuring alignment with client-specific business requirements.

Oracle E-Business Suite (EBS) Project | *Kasrawy Automotive Group* 2026

- Delivered custom and standard reports modification for Kasrawy Automotive Group on Oracle E-Business Suite (EBS).
- Updated and enhanced custom modules built on OAF (Oracle Application Framework) pages, ensuring alignment with client-specific business requirements.

Multi-Agent HR Absence Assistant — Oracle Fusion | *Infrarose* 2026

- Designed a two-agent orchestrated architecture on Oracle Fusion AI Agent Studio: an absence-management agent handling creation/retrieval against Oracle Fusion HCM REST APIs, and a policy RAG agent answering leave-policy questions.
- Built pre-validation logic checking an employee's existing absence records before submission to prevent overlapping or duplicate entries (resolving conflict error ANC-3405005).
- Delivered a coordinating orchestration layer routing each query to the correct agent, presenting a single unified conversational interface, and evolved the design to a supervisor/worker pattern for a manager demo.

Bilingual RAG System (Arabic/English) | GitHub 01/2026 – Present

- Architected a production-grade RAG API supporting Arabic and English with **902ms average end-to-end latency** via fully async FastAPI and asyncpg, with a provider-agnostic LLM interface supporting OpenAI GPT and Cohere Command-R+.
- Designed a dual vector database layer (Qdrant + PgVector HNSW) using a factory pattern, enabling zero-code provider switching through environment configuration alone.

- Built a parallelized OCR pipeline (ThreadPoolExecutor + Mistral OCR + Gemini Vision) to process scanned PDFs and mixed Arabic/English documents at scale.
- Deployed to Microsoft Azure with CI/CD via GitHub Actions; instrumented **8 Docker Compose services** behind Nginx with Prometheus and Grafana for full production observability.
- Managed PostgreSQL schema lifecycle with Alembic migrations across three relational tables with JSONB metadata and indexed foreign keys, ensuring efficient chunk retrieval at scale.

ArabicLLM — Arabic NLP Fine-Tuning | GitHub 02/2026 – Present

- Fine-tuned Qwen2.5-1.5B on Arabic NLP tasks (NER, summarization, keyword extraction) using LoRA rank 64 via LLaMA-Factory (SFT), matching GPT-4o-level structured output quality at zero per-request inference cost through full self-hosting.
- Built a knowledge distillation pipeline using GPT-4o as teacher model to generate high-quality structured training data from raw Arabic news stories, validated with Pydantic v2 schemas.
- Deployed a production vLLM 0.6.6 inference server with LoRA adapter hot-loading on an NVIDIA T4 GPU, sustaining **20 concurrent users** at **14s average latency** at 4096-token context, validated via Locust load testing.
- Built a live Streamlit inference UI with real-time token streaming over an OpenAI-compatible API; tracked 3 training epochs at learning rate 2e-4 with Weights & Biases (W&B).